LOGage 2026

OMNI-CHANNEL DISTRIBUTION SYSTEM

Competition Examination Paper | Round 2
Language: English

1. INPUT DATA (DATASETS)

Contestants are provided with three partially-cleaned data files:

File 01	SKU Master Data - product dimensions (L/W/H), weight, packaging specifications (PCS/CTN/Pallet), and volume (CBM).

File 02	Transaction Log (My Phuoc & Vinh Loc Warehouses) - outbound shipment history from June 2025 to December 2025, including SKU code, quantity, ship-to point, order creation datetime, and total CBM.

File 03	Distributor Network - list of distributors with delivery coordinates and service regions.

2. EXAMINATION QUESTIONS

PART 1: FLOW ANALYSIS & CUSTOMER PROFILING (NETWORK PROFILING)

Question 1.1. ABC-XYZ Analysis

Perform an ABC analysis of all SKUs based on two dimensions:

- Quantity (outbound volume in the last 6 months of 2025)
- Order Frequency (number of times each SKU appeared in transactions)

Your analysis must:

– Clearly define the classification thresholds used (e.g., top 20% by volume = Class A)
– Build an ABC–XYZ combined matrix if possible
– Identify the 'Fast-Moving' SKU group that creates the highest operational pressure on warehouse fulfillment

  - Quantify what percentage of total volume/frequency the Fast-Moving group accounts for

Question 1.2. Distribution Heatmap

Using the Transaction Log data:

- Aggregate outbound volume and order count by Ship-to region
- Visualize the geographic distribution of orders (map, heatmap, or bar chart by region)
- Identify the top customer clusters with the highest order density
- Highlight any significant imbalance between the two warehouse locations (My Phuoc vs. Vinh Loc)

Question 1.3. Order Profile Analysis

Compare the order profiles of two distinct customer segments:

- Modern Trade (Co.op Mart, Lotte, Big C, etc.) - large-format retail
- Traditional Trade / Small Distributors - smaller, fragmented orders

Your comparison should cover at least three of the following dimensions: average order size (CBM/pcs), order frequency, SKU breadth per order, lead time sensitivity, packaging unit (CTN vs. pallet), geographic spread.

PART 2: OMNI-CHANNEL FULFILLMENT STRATEGY DESIGN

Question 2.1. Network Model Evaluation & Design

Currently, outbound orders are split between two Regional Distribution Centers (RDCs):

- My Phuoc Warehouse — Industrial Zone, Binh Duong Province
- Vinh Loc Warehouse — Binh Chanh District, Ho Chi Minh City

Tasks:

1. **Evaluate the current two-RDC model: identify its strengths and limitations for serving both B2B and B2C channels simultaneously.**
2. **The brand intends to launch a B2C / E-commerce channel with a 2–4 hour delivery SLA across Ho Chi Minh City. Assess whether the existing RDC structure is sufficient, or whether additional 'dark store' / micro-fulfillment nodes inside the city are necessary.**
3. **Support your recommendation with data: calculate the straight-line (or road) distance from each warehouse to major HCMC districts, cross-reference with order density data from Question 1.2, and propose specific node locations if urban nodes are recommended.**

Question 2.2. Inventory Optimization

Tasks:

4. **Calculate the Safety Stock for all Class A SKUs (identified in Question 1.1) using demand variability data from the 6-month transaction history. Show your formula, inputs, and results in a table.**
5. **Propose an Inventory Pooling strategy that allows stock to be shared dynamically between the B2B channel and the B2C channel. Address: how to prevent simultaneous stockout in one channel while the other holds excess inventory, and what system logic or rules govern the allocation.**

PART 3: WAREHOUSE OPERATIONS SOLUTIONS

Question 3.1. Slotting Optimization

Using the ABC classification from Part 1 and the SKU dimensions from the Master Data:

- Design a slotting plan that assigns SKUs to the most appropriate storage zone
- Class A / Fast-Moving SKUs → Pick-face zone (lower racks, ergonomic height, close to packing area)
- Class B SKUs → Intermediate zone with replenishment logic
- Class C / Slow-Moving SKUs → Reserve zone (upper racks, bulk storage)

Demonstrate quantitatively how your slotting plan achieves at least a 30% reduction in picker travel time compared to a random storage baseline. Use SKU weight, size, and velocity as inputs.

Question 3.2. Omni-Channel Pick-and-Pack Process Design

Design a unified pick-and-pack process for the following two order types:

- **B2B Orders (large volume, few SKUs): Apply a Pallet / Total Picking model. Pickers fulfill entire pallets or large CTN quantities per order in a single pass.**
- **B2C / E-commerce Orders (small quantity, many SKUs, multiple concurrent orders): Apply a Batch Picking + Zone Routing model. Group multiple small orders into a single pick wave; sort at packing.**

Mandatory deliverable: Draw a Flowchart that shows the complete decision logic, including the specific scenario where the same SKU is ordered simultaneously by both a B2B client and a B2C customer. How is the available stock allocated, and what escalation rules apply if stock is insufficient for both?